

ROLJOHN FRILLES

Quezon City, Philippines • psychammer87@gmail.com • 0954 176 0422 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

WORK EXPERIENCE

Department of Science and Technology (DOST-CO-PES-ITD)

2025

Software Developer Intern | Taguig City, Philippines

- Developed full-stack library modules with server-side data handling, validation, and interactive UIs for the Balik Scientist Program using Laravel.
- Engineered a custom WordPress plugin for the Science, Technology, and Innovation Statistics (STIS) data management system.
- Built a Vue.js front-end for complex data entry, integrating Handsontable.js for spreadsheet-style manipulation and Chart.js for real-time visualization.
- Developed PHP backend functions and WordPress AJAX endpoints to securely handle data saving and retrieval.

PROJECTS

HydroCast: Marikina River Basin Flood Forecasting

2026

- Engineered a hybrid deep learning architecture (STGAE, GAT, Transformer) to forecast river water levels up to 48 hours ahead using 13 years of hydrometeorological data.
- Handled 68.9% missing real-world sensor data via unsupervised pre-training for signal imputation.
- Deployed the inference engine as a cloud-ready FastAPI/Docker microservice, decoupling ML processing from the mobile frontend.

Kuwartrack: Personal Finance Manager

2025

- Led a team building a financial tracking mobile app with a weekly expense dashboard, budget-setting transaction module, and category breakdown analytics.

Smart Logistics Optimizer (Knapsack Algorithm)

2024

- Built a mobile logistics app applying the knapsack algorithm to optimize truck/pallet loading for space utilization and profit margins.

EDUCATION

Polytechnic University of the Philippines - Manila

Oct 2022 - Present

B.S. in Computer Science | Manila, Philippines | Completed Academic Requirements; Graduation Scheduled Sept. 2026

- Consistent President's Lister
- DOST-SEI Scholar

LEADERSHIP & CO-CURRICULAR ACTIVITIES

DEVCON Manila Chapter — Technical Speaker, Push2Prod 2026

Jun 2026

- Delivered a 25-minute technical talk, "Preparing for the Real World: Testing, Improving, and Deploying Your Project," to 150+ IT/CS students, covering deployment of ML architectures into production cloud microservices.

AWS Cloud Club Skill Building - PUP Manila — ML/AI Co-Head

2023 - 2024

- Led hands-on workshops on foundational Python for AI/ML, covering NumPy, Matplotlib, and Pandas.

Collabothon: IdeaJam — Co-Facilitator

Feb 2024

- Co-facilitated brainstorming sessions guiding participants through emerging-tech solution design.

SKILLS

Languages & Frameworks: Python, JavaScript, HTML/CSS, PHP, Flutter, Laravel

Data Science & ML: PyTorch, NumPy, Pandas, Matplotlib, Neural Networks, Regression, Classification

Backend & Deployment: FastAPI, Docker, Render, Gradio

CERTIFICATIONS & PUBLICATIONS

- Computer Vision: Deep Learning with PyTorch — Object Localization & Image Segmentation, Coursera Project Network (2023)
- "[My First Exposure to Machine Learning with PyTorch](#)," Tutorials Dojo — Technical Author (Jun 2026)